

Human Review: automorphism-map noninjectivity

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Reviewed for survey inclusion, but not independently checked in detail by the human reviewer; excluded for the time being.

Verdict: Exclude from the verified successful examples at this stage.

Human Comments

The proposed counterexample is an elementary observation about the literal formulation of the question. It argues as follows. If X is a Banach space and P is a nonzero finite-rank projection, then

$$U = I + P$$

is invertible, and conjugation

$$\text{Ad}_U(T) = UTU^{-1}$$

defines an automorphism of $\mathcal{L}(X)$. Since $U - I = P$ is compact, this automorphism induces the identity map on the Calkin algebra

$$\text{Cal}(X) = \mathcal{L}(X)/\mathcal{K}(X).$$

The packet further argues that Ad_U need not be the identity automorphism of $\mathcal{L}(X)$. Thus, if this elementary calculation is accepted, then the literal map from automorphisms of $\mathcal{L}(X)$ to the induced automorphisms of the Calkin algebra would not be injective.

However, I do not think this should be included as a successful project example for the time being. The source manuscript itself raises several reliability concerns. It appears to claim a large number of substantial results at once, including realization, rigidity, ideal classification, and new reflexive Calkin-algebra examples. Some parts of the manuscript also look internally suspicious. For example, the running header refers to “Strongly Singular Kirchhoff-Type Equations”, which is unrelated to the paper title and content. More substantively, the paper appears to pass from finite direct sums of copies of $C(K)$ to matrix algebras $M_m(C(K))$ by using a linear identification, even though these are not isomorphic as Banach algebras for $m > 1$ since the former is commutative and the latter is not. There is also a claim involving ℓ_2 with pointwise multiplication as a Calkin algebra, which is problematic since such an algebra is not unital.

There is a second issue with the formulation of the relevant open question. The remark asks about injectivity of a natural map to $\text{Homeo}(K)$, but it also phrases the issue in terms of possible non-inner automorphisms in the kernel. The candidate counterexample only supplies an inner automorphism, coming from conjugation by an invertible compact perturbation of the identity. Therefore it answers the literal wording, but probably not the more meaningful intended question if inner automorphisms were supposed to be ignored.

Short Version

The finite-rank conjugation argument is a possible elementary objection to the literal wording: inner automorphisms implemented by invertible compact perturbations of the identity should be invisible in the Calkin algebra. But the argument has not been independently checked in detail by the human reviewer, the source paper appears unreliable enough, and the intended question may differ enough from the literal wording that this packet should be excluded for now.

Literature and Follow-Up

This example should not be used as evidence of mathematical progress unless the source manuscript is independently validated or the authors clarify that the literal inner-automorphism obstruction is genuinely relevant to their intended question. If the intended problem concerns outer automorphisms or automorphisms modulo inner automorphisms, then the present packet does not address it.

Review Outcome

Exclude this packet from the verified successful examples for the time being. It may be retained as a cautionary case: the automatic system found a possible elementary objection to the literal wording, but the argument has not been independently checked in detail by the human reviewer, and the surrounding source and likely intended question make it unsuitable for inclusion as a positive mathematical-discovery example.